

Precision Aerospace Manufacturing: Controlling Variation, Measurement Risk & Component Reliability

Company: **Rolls-Royce (Illustrative Scenario)**
Theme: **Advanced Aerospace Manufacturing QE**

Prepared by: **Jagriti Singh | Quality Engineer**
Date: **26 September 2026** | Ref ID: **QE-61**

1. CASE CONTEXT & PROBLEM OVERVIEW

This technical case study examines advanced quality engineering methodologies applied to critical precision aerospace engine components. Performance is influenced by material/supplier variation, precision machining instability, dimensional variation, measurement uncertainty, and component non-conformance.

Supplier / Material → Manufacturing Process → Measurement & Inspection → Component Quality → Engine Reliability

Quality Problem Definition: A precision aerospace machining process experiences variation in critical characteristics. The investigation determines whether variation originates from material, process, measurement, or variable interactions. **Objective:** Detect → Characterise → Prioritise → Isolate → Correct → Validate → Control.

2. MEASUREMENT SYSTEM ANALYSIS (MSA)

Positioned before process capability evaluations to determine if variation stems from the process or measurement error.

- **Repeatability & Reproducibility (GR&R):** Equipment and operator variation checks.
- **Bias, Stability & Resolution:** Calibration offset, temporal drift, and discrimination.

Observed Variation → MSA Check → Separate Gauge Error

Note: "Capability conclusions are only meaningful when the measurement system is sufficiently capable."

3. PROCESS CAPABILITY ANALYSIS

Evaluates whether a stable process consistently meets specified requirements after confirming measurement reliability.

- **Stability Check:** Verifying absence of special causes via control charts.
- **Capability Assessment:** Quantifying spread relative to limits (Cp, Cpk conceptually).

Distinction: **Process Stability** ensures predictable operation over time, whereas **Process Capability** evaluates whether predictable spread fits within engineering tolerances.

4. ROOT-CAUSE ANALYSIS (FISHBONE & 5 WHY)

Structured investigation to uncover underlying mechanical and operational mechanisms driving non-conformances.

- **Fishbone (Ishikawa):** Structures contributors across 6M: *Man, Machine, Method, Material, Measurement, Environment*.
- **5 Why Analysis:** Deepens investigation from symptom to root physical/systemic cause.

Symptom → 6M Causes → Evidence → Root Cause

5. RISK & FAILURE-MODE ANALYSIS (FMEA)

Systematic risk evaluation differentiating design and manufacturing risks to prioritize engineering actions.

- **Design FMEA:** Evaluates potential failure modes at system/ design level.
- **Process FMEA:** Evaluates how manufacturing steps create or transfer risk.

RPN: $Severity \times Occurrence \times Detection = RPN$

Note: RPN is a prioritisation aid; safety criticality and customer impact must guide decisions.

ROLLS-ROYCE — ADVANCED AEROSPACE MANUFACTURING QUALITY ENGINEERING

100 DAYS TQM & QUALITY ENGINEERING CHALLENGE — DAY 61/100

Precision Aerospace Manufacturing: Dominant Cause, Experimentation & Control

Company: **Rolls-Royce (Illustrative Scenario)**
Theme: **Advanced Aerospace Manufacturing QE**

Prepared by: **Jagriti Singh | Quality Engineer**
Date: **26 September 2026** | Ref ID: **QE-61**

6. DOMINANT CAUSE & EXPERIMENTATION

Shainin Red X: Moving from many variables to candidate variables via controlled comparisons, isolating the dominant variable driving variation.



Controlled Experimentation: Establishing strict hypotheses and comparative trials to confirm cause-and-effect before standardization.

7. PREVENTION & IMPROVEMENT

Poka-Yoke (Error-Proofing): Implementing physical/logic interlocks for setup, part orientation, and automated parameter inspection.

Kaizen: Continuous incremental improvement of work methods, process controls, inspection flow, and standardization.

8. COMPLETE QUALITY ENGINEERING FLOW FRAMEWORK

Tool / Stage	Core Engineering Purpose
MSA	Validate measurement system reliability and separate gauge error from process variation.
Process Capability	Quantify process performance relative to engineering specifications (Cp, Cpk).
Fishbone & 5 Why	Structure potential causes across 6M categories and deepen investigation to root mechanisms.
FMEA (Design / Process)	Assess failure risks, evaluate severity/occurrence/detection, and calculate RPN.
Shainin Red X	Isolate the single dominant variable causing the majority of variation.
Experimentation	Conduct controlled statistical testing to confirm verified cause-and-effect.
Poka-Yoke & Kaizen	Error-proof operational steps and drive continuous incremental process refinement.
PDCA & Pilot	Implement structured Plan-Do-Check-Act cycles and test pilot changes safely.
Validation & SPC	Verify improvement effectiveness and monitor ongoing process stability via control charts.
Control Plan	Sustain long-term gains through standardized reaction plans, frequencies, and ownership.

9. PDCA, PILOT & PROCESS CONTROL

- **PLAN:** Define problem, evidence, and targets.
- **DO:** Execute controlled pilot implementation.
- **CHECK:** Analyse post-implementation evidence.
- **ACT:** Standardize validated improvements.

SPC & Control Plan: Monitor critical characteristics via control charts, supported by reaction plans and defined frequencies.

10. OUTCOMES & KEY INSIGHT

- Improved measurement confidence & reduced variation.
- Enhanced process capability & defect prevention.
- Robust risk prioritization & sustainable controls.

"In aerospace manufacturing, controlling quality requires more than detecting defects — measuring correctly, understanding variation, isolating dominant causes, validating improvements, and maintaining control over critical characteristics."

Disclaimer: Illustrative case study independently developed by Jagriti Singh as part of the 100 Days TQM & Quality Engineering Challenge. This is not an official Rolls-Royce project, internal Rolls-Royce case study, or representation of Rolls-Royce proprietary processes, systems, data or results.